

A NEW THEORY OF CREDIT LINES (WITH EVIDENCE)

Jason Roderick Donaldson

Naz Koont

Giorgia Piacentino

Victoria Vanasco

FACTS

Credit lines make up c. \$2T of committed credit, bulk of bank credit (80%)
Berg–Saunders–Steffen 20, Greenwald–Kraimer–Paul 21, Chodorow-Reich
et al. 21, and Sufi 09

Credit lines are rarely drawn (6%) even in crises (24%)
Ivashina–Scharfstein 10, Greenwald 21

Credit lines are bundled with loans, especially to risky firms (80%)
See below

Credit lines are sometimes revoked by lenders
Falato–Chodorow-Reich 22

QUESTIONS

Q1. Why are credit lines so common, even if rarely used?

Q2. Why are credit lines bundled with loans?

THIS PAPER

Dynamic model of borrower B issuing debt

Admati et al 17/DeMarzo–He 21

Friction: Non-exclusivity

After borrowing from one lender at t , borrows from another at $t + dt$

Innovation: Allow for credit lines (CLs)

Options to borrow $\tilde{p}d\tilde{Q}$ against face value $d\tilde{Q}$

RESULTS

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (ration Q so $p > MC$)

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (rations Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (rations Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

R1. New role of credit lines: Commitment device not to dilute

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (ration Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

R1. New role of credit lines: Commitment device not to dilute

New lenders don't lend if anticipate credit line drawn (diluting them)

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (ration Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

R1. New role of credit lines: Commitment device not to dilute

New lenders don't lend if anticipate credit line drawn (diluting them)

R2. New role of bundling: Pairing loan with CL commits to optimal debt

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (rationes Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

R1. New role of credit lines: Commitment device not to dilute

New lenders don't lend if anticipate credit line drawn (diluting them)

R2. New role of bundling: Pairing loan with CL commits to optimal debt

R3. New role of lender commitment: Revocation $\uparrow$ borrower debt

RESULTS

BM1. Exclusive comp.: B acts as static monopolist (rationes Q so $p > MC$)

BM2. No credit lines: Gains from trade lost to dilution (DeMarzo–He 21)

R1. New role of credit lines: Commitment device not to dilute

New lenders don't lend if anticipate credit line drawn (diluting them)

R2. New role of bundling: Pairing loan with CL commits to optimal debt

R3. New role of lender commitment: Revocation $\uparrow$ borrower debt

T. Empirical evidence: two natural experiments + cross-sectional tests

MODEL

OVERVIEW

Infinite-horizon sequential borrowing: B borrows from one lender at each t

Credit line-debt bundle at date 0 and new debt afterward

B's cost of debt $c(Q_t)$ increasing and concave in stock of debt Q_t

Captures expected coupon payment $\downarrow$ as default prob. $\uparrow$ in Q

Universal risk-neutrality, deep pockets, and discounting at rate ρ

NB: No revocation; no risk; no CLs at $t > 0$

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

Q_t : stock of debt

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

Q_t : stock of debt

dQ_t : new debt issued

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

Q_t : stock of debt

dQ_t : new debt issued

$c(Q_t)$: cost of debt with $c(0) = 0$, $c' > \bar{c}' > 0$ and $c'' < 0$

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

Q_t : stock of debt

dQ_t : new debt issued

$c(Q_t)$: cost of debt with $c(0) = 0$, $c' > \bar{c}' > 0$ and $c'' < 0$

p_t : unit price of debt

B's VALUE

B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

dt : time increment

$y dt$: cash flow

Q_t : stock of debt

dQ_t : new debt issued

$c(Q_t)$: cost of debt with $c(0) = 0$, $c' > \bar{c}' > 0$ and $c'' < 0$

p_t : unit price of debt

B's value: $V_t = \int_0^\infty e^{-\rho s} v_{t+s} ds$

LENDERS' VALUE

Lenders' flow payoff from unit debt given stock Q_t : $\gamma(Q_t)$

LENDERS' VALUE

Lenders' flow payoff from unit debt given stock Q_t : $\gamma(Q_t)$

Value of stream of coupons on unit of debt:

$$\Gamma_t = \int_0^\infty e^{-\rho s} \gamma(Q_{t+s}) ds$$

CONTRACTS

Loans (p_t, dQ_t) : Borrow $p_t dQ_t$ against face value dQ_t

CONTRACTS

Loans (p_t, dQ_t) : Borrow $p_t dQ_t$ against face value dQ_t

Credit lines $(\tilde{p}, d\tilde{Q})$: Option to borrow $\tilde{p} d\tilde{Q}$ against face value $d\tilde{Q}$

CONTRACTS

Loans (p_t, dQ_t) : Borrow $p_t dQ_t$ against face value dQ_t

Credit lines $(\tilde{p}, d\tilde{Q})$: Option to borrow $\tilde{p} d\tilde{Q}$ against face value $d\tilde{Q}$

($d\tilde{Q}$ put options on debt with strike $\tilde{p}$)

ASSUMPTIONS

A1: Dilution: $\gamma' < 0$ and $\lim_{Q \rightarrow \infty} \gamma(Q) = 0$

A2: Gains from trade:

Positive at $Q = 0$: $\gamma(0) > c'(0)$

Negative as $Q \rightarrow \infty$: $\gamma(\infty) < c'(\infty)$

CORPORATE FINANCE INTERPRETATION

c flexible way to capture repayment risk and agency frictions

CORPORATE FINANCE INTERPRETATION

c flexible way to capture repayment risk and agency frictions

Let π be default prob., δ coupon rate, τ_c corporate tax, $k > \delta$ default costs:

$$\gamma(Q) = (1 - \pi(Q))\delta$$

$$c(Q) = \underbrace{(1 - \pi(Q))\delta Q}_{E[\text{coupons}] = \gamma(Q)Q} - \underbrace{\tau_c \delta Q}_{\text{tax shield}} + \underbrace{\pi(Q)Qk}_{E[\text{default costs}]}$$

CORPORATE FINANCE INTERPRETATION

c flexible way to capture repayment risk and agency frictions

Let π be default prob., δ coupon rate, τ_c corporate tax, $k > \delta$ default costs:

$$\gamma(Q) = (1 - \pi(Q))\delta$$

$$c(Q) = \underbrace{(1 - \pi(Q))\delta Q}_{E[\text{coupons}] = \gamma(Q)Q} - \underbrace{\tau_c \delta Q}_{\text{tax shield}} + \underbrace{\pi(Q)Qk}_{E[\text{default costs}]}$$

Baseline assumptions easy to satisfy

CORPORATE FINANCE INTERPRETATION

c flexible way to capture repayment risk and agency frictions

Let π be default prob., δ coupon rate, τ_c corporate tax, $k > \delta$ default costs:

$$\gamma(Q) = (1 - \pi(Q))\delta$$

$$c(Q) = \underbrace{(1 - \pi(Q))\delta Q}_{E[\text{coupons}] = \gamma(Q)Q} - \underbrace{\tau_c \delta Q}_{\text{tax shield}} + \underbrace{\pi(Q)Qk}_{E[\text{default costs}]}$$

Baseline assumptions easy to satisfy:

1. c increasing: $(\pi(Q)Q)' > 0$
2. c concave: $(\pi(Q)Q)'' < 0$
3. c' bounded above zero: $\lim_{Q \rightarrow \infty} (\pi(Q)Q)' > \bar{c}'_{lb} > 0$

SOLUTION CONCEPT: MARKOV PERFECT EQ.

At each date, lenders offer bundles/loans and B chooses one offer s.t.

B and lenders max expected future lifetime utility given beliefs

Beliefs are consistent

B's balance sheet $(Q_t, (\tilde{p}, d\tilde{Q}))$ is the state variable

SOLUTION CONCEPT: MARKOV PERFECT EQ.

At each date, lenders offer bundles/loans and B chooses one offer s.t.

B and lenders max expected future lifetime utility given beliefs

Beliefs are consistent

B's balance sheet $(Q_t, (\tilde{p}, d\tilde{Q}))$ is the state variable

NB: For a given CL, draw down is binary action

SOLUTION CONCEPT: MARKOV PERFECT EQ.

At each date, lenders offer bundles/loans and B chooses one offer s.t.

B and lenders max expected future lifetime utility given beliefs

Beliefs are consistent

B's balance sheet $(Q_t, (\tilde{p}, d\tilde{Q}))$ is the state variable

NB: For a given CL, draw down is binary action

Notation: $\tilde{V}$ value function with credit line; V without (after draw down)

BENCHMARKS

BENCHMARK 1: EXCLUSIVE COMPETITION

BENCHMARK 1: EXCLUSIVE COMPETITION

If B commits to one lender (of those competing at date 0) forever

B borrows at Date 0 and never again

Price is above marginal cost

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

Balance costs (e.g. distress) with gains from trade (e.g. tax shield)

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

Balance costs (e.g. distress) with gains from trade (e.g. tax shield)

Durable goods monopolist intuition:

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

Balance costs (e.g. distress) with gains from trade (e.g. tax shield)

Durable goods monopolist intuition: $MC = MR$

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

Balance costs (e.g. distress) with gains from trade (e.g. tax shield)

Durable goods monopolist intuition: $MC = MR = (pQ)'$

EXCLUSIVE COMP: INTUITION

Corporate finance intuition: Static trade-off theory

Balance costs (e.g. distress) with gains from trade (e.g. tax shield)

Durable goods monopolist intuition: $MC = MR = (pQ)'$

Take effect of Q on p into account $\implies p > MC$

$\implies p > MC$: price higher/quantity lower than with perfect competition

BENCHMARK 2: NO CREDIT LINES (& NON-EXCLUSIVITY)

BENCHMARK 2: NO CREDIT LINES

BENCHMARK 2: NO CREDIT LINES

Absent credit lines, with $dt \rightarrow 0$

B issues debt continuously

B captures no surplus

Price equals marginal cost

BENCHMARK 2: NO CREDIT LINES

Absent credit lines, with $dt \rightarrow 0$

B issues debt continuously

B captures no surplus

Price equals marginal cost

Cf. Leverage ratchet effect and Coase Conjecture
(Admati et al. 18, DeMarzo–He 21, Coase 72,...)

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in q

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \ (\implies q > 0)$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus:

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho}$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho} \implies V(0) = \frac{y - c(0)}{\rho}$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho} \implies V(0) = \frac{y - c(0)}{\rho}$ as if $q_t \equiv 0$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho} \implies V(0) = \frac{y - c(0)}{\rho}$ as if $q_t \equiv 0$

Pricing:

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho} \implies V(0) = \frac{y - c(0)}{\rho}$ as if $q_t \equiv 0$

Pricing: Using expressions for V' and V above: $-V'(Q) = p(Q) = \frac{c'(Q)}{\rho}$

NO CREDIT LINES: PROOF

As $dt \rightarrow 0$, HJB for B's value V : $\rho V(Q) = y + p(Q)q - c(Q) + V'(Q)q$

Issuance: Linear in $q \implies$ coefficient $p(Q) + V'(Q) = 0 \implies q > 0$

Surplus: HJB becomes $V(Q) = \frac{y - c(Q)}{\rho} \implies V(0) = \frac{y - c(0)}{\rho}$ as if $q_t \equiv 0$

Pricing: Using expressions for V' and V above: $-V'(Q) = p(Q) = \frac{c'(Q)}{\rho}$

NB: c concave $\implies V$ convex ($V'' = -c''/\rho > 0$)

NO CREDIT LINES: INTUITION

Corporate finance intuition: Costs of new debt born by existing debt

NO CREDIT LINES: INTUITION

Corporate finance intuition: Costs of new debt born by existing debt

Debt in place $\implies$ new debt tempting $\implies$ “ratchet effect” eats surplus

RESULTS

LEMMA: RATCHET EFFECT FOR CREDIT LINES

LEMMA: RATCHET EFFECT FOR CREDIT LINES

Let B have CL $(\tilde{p}, d\tilde{Q})$ in place s.t. $\sim$ between drawing or staying at Q_0

B weakly prefers to draw for $Q > Q_0$

RATCHET EFFECT OF CLs: INTUITION

c concave $\implies$ more debt B has, less costly to have more

RATCHET EFFECT OF CLs: INTUITION

c concave $\implies$ more debt B has, less costly to have more

Idea: Higher $Q \implies$ lower repayment prob. $\implies$ lower cost of $d\tilde{Q}$

RATCHET EFFECT OF CLs: INTUITION

c concave $\implies$ more debt B has, less costly to have more

Idea: Higher $Q \implies$ lower repayment prob. $\implies$ lower cost of $d\tilde{Q}$

Akin to leverage ratchet effect: Higher debt begets higher debt

R1: RATCHET-ANTI-RATCHET EFFECT

R1: RATCHET-ANTI-RATCHET EFFECT

Let $(\tilde{p}, d\tilde{Q})$ be CL s.t. $B \sim$ drawing at Q_0

B doesn't take any new debt ($dQ_t \equiv 0$)

R1: RATCHET-ANTI-RATCHET EFFECT: PROOF

Step 1: Find price $\underline{p}$ s.t. B borrows dQ (and thus draws CL) if $p > \underline{p}$

R1: RATCHET-ANTI-RATCHET EFFECT: PROOF

Step 1: Find price $\underline{p}$ s.t. B borrows dQ (and thus draws CL) if $p > \underline{p}$

Step 2: Find price $\bar{p}$ s.t. lender lends dQ if $p < \bar{p}$

R1: RATCHET-ANTI-RATCHET EFFECT: PROOF

Step 1: Find price $\underline{p}$ s.t. B borrows dQ (and thus draws CL) if $p > \underline{p}$

Step 2: Find price $\bar{p}$ s.t. lender lends dQ if $p < \bar{p}$

Step 3: Show $\underline{p} > \bar{p}$ for $d\tilde{Q}$ large enough

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

Anticipating dilution lowers price lenders' willing to pay

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

Anticipating dilution lowers price lenders' willing to pay

Price so low that B doesn't want to borrow

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

Anticipating dilution lowers price lenders' willing to pay

Price so low that B doesn't want to borrow

$\implies$ credit lines implement exclusivity absent borrower commitment

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

Anticipating dilution lowers price lenders' willing to pay

Price so low that B doesn't want to borrow

$\implies$ credit lines implement exclusivity absent borrower commitment

NB: Self-enforcing independent of who holds debt

R1: RATCHET-ANTI-RATCHET: INTUITION

When B borrows from new lenders, draws on credit line diluting them

Anticipating dilution lowers price lenders' willing to pay

Price so low that B doesn't want to borrow

$\implies$ credit lines implement exclusivity absent borrower commitment

NB: Self-enforcing independent of who holds debt

Debt can be traded, as in practice, unlike in monitoring-based theories

R2: BUNDLING

R2: BUNDLING

At date 0, B chooses a bundle of a loan (p, Q_0) and CL $(\tilde{p}, d\tilde{Q})$ s.t.

(p, Q_0) coincides with outcome of exclusive competition

$(\tilde{p}, d\tilde{Q})$ makes B indifferent to drawing at Q_0 with $d\tilde{Q}$ large

R2: BUNDLING: PROOF

Recall that under candidate bundle

B never borrows again by R1

B's payoff maximized s.t. lenders break even by BM1

R2: BUNDLING: PROOF

Recall that under candidate bundle

B never borrows again by R1

B's payoff maximized s.t. lenders break even by BM1

$\implies$ At any other bundle another lender can profitably enter

R2: BUNDLING: PROOF

Recall that under candidate bundle

B never borrows again by R1

B's payoff maximized s.t. lenders break even by BM1

$\implies$ At any other bundle another lender can profitably enter

Can post candidate with price p replaced by $p + \epsilon$

R2: BUNDLING: INTUITION

Credit lines allow B to commit not to dilute by R1

Use credit lines to implement optimum without dilution (i.e. BM1)

REVOCATION

So far: Credit lines always honored if drawn

Now: Credit lines honored with prob. α (else disappear) if drawn

REVOCATION

So far: Credit lines always honored if drawn

Now: Credit lines honored with prob. α (else disappear) if drawn

$\implies$ Payoffs given drawing at Q_0 convex comb. of those at $Q_0 + d\tilde{Q}$ and Q_0

R3: LENDER REVOCATION RISK INCREASES B's DEBT

R3: REVOCATION RISK

With revocation risk $1 - \alpha$, B can sustain debt Q_0 to limit for $dt \rightarrow 0$:

$$\Gamma(Q_0) - \frac{c'(Q_0)}{\rho} \leq \frac{\alpha}{1 - \alpha} \frac{c'(\infty)}{\rho}$$

R3: REVOCATION RISK: INTUITION

Credit line revocation risk makes new lenders willing to lend at higher p

Less worried about dilution by credit line if might not be there

R3: REVOCATION RISK: INTUITION

Credit line revocation risk makes new lenders willing to lend at higher p

Less worried about dilution by credit line if might not be there

$\implies$ harder to deter new debt

R3: REVOCATION RISK: INTUITION

Credit line revocation risk makes new lenders willing to lend at higher p

Less worried about dilution by credit line if might not be there

$\implies$ harder to deter new debt

$\implies$ harder to commit not to dilute in the first place

PREDICTIONS & TESTS

PREDICTION 1: BUNDLING

Prediction: Credit lines are bundled with loans

Reason: CL protects loan from dilution

Evidence: > 70% term loans bundled with CL at origination

PREDICTION 2: BUNDLING PROPENSITY

Prediction: Bundling propensity higher if dilution risk higher

Reason: Protection valuable only if dilution likely

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 3: SIZE

Prediction: Credit lines are large

Reason: CL deters new borrowing only if large

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 3: SIZE

Prediction: Credit lines are large

Reason: CL deters new borrowing only if large

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 4: SIZE IN CROSS-SECTION

Prediction: Credit lines are larger if dilution risk higher

Reason: Deterrant more valuable when dilution more likely

Evidence: Larger CLs if default likely or lenders less concentrated

PREDICTION 5: DRAW DOWNS

Prediction: Credit lines are rarely drawn

Reason: CL off-equilibrium threat

Evidence: 45% of borrowers utilize none of CL; median 17%

PREDICTION 6: BORROWING

Prediction: Credit lines deter subsequent borrowing

Reason: Anti-ratchet

Evidence: Increased prob. CL present increased borrowing

PREDICTION 6: BORROWING

Prediction: Credit lines deter subsequent borrowing

Reason: Anti-ratchet

Evidence: Increased prob. CL present increased borrowing

Experiments: Changes in revocation prob. and regulatory cost of CL

PREDICTION 7: RATES

Prediction: Rates on CLs relatively low

Reason: Rate s.t. lender worse off when lends

Evidence: Median rate on CL $\approx$ that on the associated term loan

CONTRIBUTIONS

CONTRIBUTIONS

Credit lines:

In lit.: Insurance to take on more debt

Here: Commitment to take on less debt

CONTRIBUTIONS

Credit lines:

In lit.: Insurance to take on more debt

Here: Commitment to take on less debt

Ratchet effects in dynamic corporate finance:

In lit.: Excessive debt and zero surplus

Here: Allowing for credit lines makes ratchet effect self-deterring

CONTRIBUTIONS

Credit lines:

In lit.: Insurance to take on more debt

Here: Commitment to take on less debt

Ratchet effects in dynamic corporate finance:

In lit.: Excessive debt and zero surplus

Here: Allowing for credit lines makes ratchet effect self-deterring

Latent contracts:

In lit.: Help lenders support collusive outcomes

Here: Help borrowers support monopolistic outcome

CONCLUSION

CONCLUSION

Heretofore unexplored role of credit lines: commit not to dilute

Curb competition with future self

Implement monopoly outcome